The combined P7 and P8 exploratory analyses show that energy consumption in the CeDInt building is influenced by both temporal patterns and environmental or occupancy-related variables.

The P7 analysis highlighted clear differences across zones and load types. Hour of day is an important factor for HVAC, Lighting and Plugs & Sockets, although both the shape and the magnitude of the profiles vary substantially between zones. Weekday and weekend effects are particularly relevant for Lighting and for some plug loads, while HVAC appears to depend more strongly on zone and seasonality.

Monthly profiles also confirm relevant seasonal patterns. HVAC shows a strong summer component, especially in the NW zone, while Lighting and Plugs & Sockets display different seasonal behaviours depending on the zone. August has a noticeable effect on several loads, but this effect is strongly zone-dependent and should therefore be treated as a specific calendar feature rather than assuming the same impact across the whole building.

The comparison between the historical period and 2025 also shows that consumption patterns are not stable over time. Some zones and load types increase, others decrease, while others remain relatively stable. In addition, the 2020 HVAC profile differs substantially from 2021–2022, but not in the same direction for all zones. This suggests that 2020 should be treated as a potentially abnormal COVID-affected period rather than assuming a single common COVID effect across the building.

The P8 analysis complements these findings by exploring the relationship between environmental variables, occupancy and energy consumption, mainly for the NW zone during 2021–2022. Luminosity shows a moderate negative correlation with Lighting energy consumption, with a correlation of approximately −0.372. This suggests that higher luminosity is generally associated with lower lighting consumption, although other variables such as time of day and occupancy may also influence this relationship.

Occupancy is also associated with different consumption patterns. Plugs & Sockets consumption is approximately 4.8% higher when presence is detected. For Lighting, however, the relationship with occupancy changes when the hour of the day is considered. This indicates that occupancy should not be analysed independently from temporal variables such as hour of day and from environmental variables such as luminosity.

Temperature shows only a weak positive linear correlation with HVAC consumption, approximately +0.174. However, the analysis by temperature intervals suggests a non-linear relationship, with the highest average HVAC consumption observed around 22–24°C. Therefore, temperature may still be a useful forecasting variable even if its simple linear correlation is weak.

Humidity shows a very weak correlation with HVAC consumption, approximately +0.059. Most humidity observations are concentrated around 30–31%, which makes it difficult to identify a clear relationship. For this reason, humidity appears to provide less explanatory information than temperature, luminosity or occupancy in the analysed data.

Insights for Forecasting

Overall, the results suggest that future forecasting models should consider a combination of temporal, spatial, environmental and occupancy-related variables.

Relevant candidate predictors include:

- zone;
- load type;
- hour of day;
- weekday and weekend;
- month and season;
- August;
- year or historical period;
- luminosity;
- temperature;
- occupancy.

The results also suggest that the relationships between these variables and energy consumption are not always simple or linear. In particular, occupancy interacts with time of day, temperature shows a non-linear relationship with HVAC consumption, and the impact of calendar effects changes across zones and load types.

For this reason, future modelling should not rely only on simple correlations. The next stages should test combinations of variables and evaluate their actual contribution to forecasting accuracy.

Limitations

The P8 environmental analysis mainly focuses on the NW zone during 2021–2022, so these results cannot automatically be generalised to the other zones or to 2025.

In addition, some environmental variables have limited variability or sensor-related limitations, especially humidity and occupancy. The P7 analysis also shows that consumption behaviour changes across zones, years and load types, which reinforces the need for careful validation.

Finally, the exploratory relationships identified in P7 and P8 should not be interpreted as causal relationships. They provide useful evidence for feature selection and further modelling, but additional analysis is required to determine which variables actually improve forecasting performance.

Final considerations

Taken together, P7 and P8 show that energy consumption is driven by a combination of temporal behaviour, zone-specific characteristics and environmental or occupancy-related factors. The strongest overall message is that a single common consumption pattern cannot be assumed for the whole building.

Therefore, the forecasting phase should account for differences between zones and load types, while also incorporating selected temporal and environmental features. This combined approach should provide a more realistic basis for developing and evaluating future energy consumption models.